Module Three Assignment

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For Milestone Two, I selected my CS 330 Computational Graphics and Visualization final project as my artifact for the Software Design and Engineering category. This project was originally created during CS 330 and is a 3D OpenGL scene built in C++, and I worked with the Scenemanager and Viewmanager. The scene includes two vehicles, a lamp post, a tree, a grass island, textured ground, lighting, materials, and basic camera functionality. I selected this artifact because it connects strongly to my computer science coursework and my future career goals in graphics programming, game development, and becoming a technical artist.

I chose this artifact for my ePortfolio because it demonstrates both technical and visual problem-solving. The project required me to organize rendering logic, work with transformations, load textures, define materials, and build a scene using reusable functions. This showed my ability to take a larger project and then break it down into smaller, more manageable, and easier-to-swallow work. Since I want to work towards becoming a technical artist, this artifact is perfect because it combines programming, rendering systems, lighting, visual design, and optimization. It's not just a basic program that gives an output, but a visual project where the code affects what the user sees on screen.

For this enhancement, I focused on improving the software design and engineering quality of the project without completely changing how the original scene works. The biggest improvements were replacing my hardcoded texture paths with a relative path system using a texture base path constant and a new load scene texture helper method. In the original version, the texture loading used full paths from my local computer and made the project harder to run on my own machine, at times, or on other machines. This new helper method reduces the repeated code as well and makes the texture loading process much easier to understand and maintain.

I also fixed a few real issues in the code, the destruction of textures, where the original code called glGenTextures instead of glDeleteTextures. This meant the program was creating a new texture handle instead of freeing the memory. This correction allows the project to delete texture resources properly. I also fixed the FindMaterial return logic. This is because in the original code, the function could or sometimes even would return true even when a material tag wasn't found. This means the wrong material tag could be sent to another shader. This change returns false and gives a warning if a material is missing.

Some more improvements were added to CreateGltexture. The original code was written into the texture IDs array without checking whether the maximum texture slot count was reached. I added a maximum texture slot to prevent the program from going beyond the array limit and, therefore, give a clearer error message if the texture limit is reached. I also improved the set shader material, so failed material lookups are handled much better without missing data. These changes made the project a lot more stable and easier to debug, as well as allowed for the scene to be expanded later.

This enhancement also allowed for better readability and maintainability. I removed a duplicate plane mesh from the prepare scene, added name constants, such as the scene's global scale, sunlight positioning, and streetlamp positioning. These changes reduced the magic hard-coded numbers and made important scene values easier to understand. I also improved my comments throughout my enhancements, so the purpose of each change is much easier to follow. I wanted to keep the changes realistic and focused instead of setting too high an expectation and feeling like I'm making a different project.

I would say that the course outcomes I planned to meet were mostly met through this enhancement. The work I believe properly supports the outcome related to using computing tools

and techniques I've learned throughout my time at SNHU, and I also improved an existing C++ graphics project in a practical, real way. It supports the software design outcome because I had to review my original code, found my weaknesses, and then decided to focus on improvements without making drastic changes to the project. The enhancements also connect to the security mindset because of the better checks and error handling, and reduced chance of missing files or invalid data causing silent issues. The narrative and code review process also supports professional communication because I explain the changes through comments, and I explain what changes were made as well as why they were made.

The biggest challenge I faced was making meaningful improvements without overcomplicating the project. I still wanted to add moonlight or make it possible to change the lighting, and I did identify several possible future improvements through the code review, such as a more data-driven rendering system and external scene data. However, for this milestone, I wanted to keep the enhancement focused on software design and engineering instead of turning the project into something else or not having work that's fully done. I learned that improving software may not mean adding new large features, but that fixing bugs and reducing repeated code, improving portability, and making the project easier to maintain will make those meaningful changes and improvements much easier.

This enhancement was an amazing experience. Being able to pinpoint my faults, as well as review and improve the code in a more professional way, allowed me to see how much I've learned. Being able to rework an old project and make it better shows the growth I've had. This makes the artifact even stronger for my ePortfolio because it shows not only that I can create a graphics project, but I can evaluate and show what Ive learned and improve the code.